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The Power of Visualization: Using NeXTA4GMNS for Trajectory Display

Cafer Avci[1¶], **Yajun Liu**[2], **and Xuesong (Simon) Zhou**[2]

1 School of Civil and Environmental Engineering, Cornell University, Ithaca, NY, 14853, USA **2** School of Sustainable Engineering and the Built Environment, Arizona State University, Tempe, AZ, 85281, USA ¶ Corresponding author

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Summary

Contemporary transportation modeling requires accurate and intuitive visualization tools to effectively interpret increasingly complex and high-dimensional data, including multi-resolution network topologies, time-varying traffic states, detailed vehicle trajectories, and large-scale simulation outputs. We present NeXTA4GMNS (NeXTA), an open-source platform designed specifically for network and trajectory visualization that fully embraces the General Modeling Network Specification (GMNS) standard (GMNS Development Team, 2020). NeXTA provides editable, multi-resolution node-link representations of transportation networks, enabling the analysis of link performance—such as travel time, traffic volume, and speed—across different time horizons. In addition to link-level metrics, it supports the visualization of vehicle trajectories, tensor-based data structures, and shortest path computations. Unlike general-purpose GIS tools, NeXTA is specifically designed for scientific transportation modeling, offering advanced capabilities for trajectory analysis, origin-destination trip and path exploration, and seamless integration with simulation frameworks such as FHWA's AMS and dynamic traffic assignment (DTA) models (Yelchuru et al., 2017).

Statement of Need

Transportation systems have become increasingly complex in recent decades, making traditional methods for analyzing traffic flow—such as manual counts, loop detectors, and periodic surveys—insufficient for capturing the dynamic behavior of modern travel patterns (Papageorgiou, 1991). These methods often lack the spatial and temporal resolution needed to support detailed modeling, validation, and policy analysis in contemporary traffic systems.

Recognizing this limitation, GMNS community has highlighted the need for standardized network representations, along with visualization platforms that can support the scale and complexity of real-world datasets (GMNS Development Team, 2020). Visualization tools that facilitate the integration of network topology, traffic dynamics, and simulation outputs are especially critical.

NeXTA was developed in response to these needs. It is a GMNS-compliant, open-source visualization platform designed for transportation network analysis and trajectory data interpretation. Rather than focusing solely on map rendering, NeXTA supports multi-resolution node-link representations and incorporates tensor-based data structures to facilitate the analysis of spatiotemporal traffic patterns grounded in established traffic flow theory (Lighthill & Whitham, 1955; Richards, 1956).

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State of the Field

Several GIS and network analysis tools exist for transportation visualization, but none provide the combination of GMNS compliance, native simulation integration, and trajectory-level analysis that NeXTA offers. General-purpose tools such as QGIS are designed for broad spatial data tasks and require extensive plugin configuration and preprocessing to support transportation-specific workflows. Dedicated microsimulation platforms such as SUMO (Behrisch et al., 2011) and VISSIM offer simulation capability but do not provide interactive, multi-resolution post-processing visualization. Tools such as TransCAD offer planning-level network analysis but are proprietary and do not natively support empirical trajectory datasets like NGSIM (Federal Highway Administration, 2007) or GMNS-formatted inputs.

Feature	NeXTA	QGIS
Network Representation	GMNS-compliant, node-link model	GIS shapefiles and layers
Trajectory Analysis	Built-in support for microscopic trajectory data (e.g., NGSIM)	Requires external plugins and preprocessing
Traffic Simulation Integration	Native support for simulation outputs (e.g., DTALite (Zhou et al., 2012), SUMO (with post-processing) (Behrisch et al., 2011))	Limited support; not transportation-specific
Network Editing	Editable, hierarchical, and multi-resolution network views	Basic vector layer editing, not traffic-centric
Tensor-Based Modeling	Supports tensor structures for trajectory processing	Not available
Usability for Transportation Practitioners	Tailored for planners, researchers, and educators in traffic modeling	General-purpose GIS users

NeXTA's utility lies in enabling users to explore traffic dynamics, examine vehicle trajectories, and modify network structures for simulation input or scenario testing. Compared to general-purpose GIS tools, NeXTA offers built-in support for transportation-specific tasks such as interpreting outputs from microscopic simulation models and editing hierarchical transportation networks.

Datasets such as the Next Generation Simulation (NGSIM) (Federal Highway Administration, 2007) provide detailed vehicle trajectories that can be difficult to analyze without specialized tools. NeXTA helps bridge this gap by enabling interactive visualization and analysis of such high-resolution data, supporting both research and practice in transportation modeling and simulation.

Software Design

Core Features and Capabilities

Multi-resolution network visualization and editing

One of NeXTA's most powerful features is its support for seamless transitions between different levels of detail. The framework handles everything from detailed lane-level analysis to broad corridor-level planning perspectives. Users can edit and analyze networks at the link level for

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microsimulation work, aggregate up to path-level analysis for route choice studies, or zoom out to zonal-level analysis for strategic planning—all within the same interface.

Tensor-based representation of trajectories

Rather than treating vehicle trajectories as simple sequences of points, NeXTA represents them as multi-dimensional tensors that capture the full richness of movement data. This formulation enables more compact data storage, efficient data manipulation, and supports advanced analytical operations compared to traditional trajectory representations. It allows us to have more efficient storage, faster manipulation, and more sophisticated analysis capabilities than traditional approaches.

Cross-resolution and multi-view trajectory visualization

NeXTA excels at showing individual vehicle trajectories unfold in real-time while simultaneously observing aggregate flow patterns across the entire network. This capability is crucial for understanding how microscopic driving behaviors aggregate into macroscopic traffic phenomena—a fundamental challenge in traffic flow theory (Gazis, 2002; Treiber & Kesting, 2013).

Integration with simulation frameworks

The platform seamlessly integrates with popular traffic simulation tools, supporting data exchange and synchronization. This integration philosophy aligns with FHWA's emphasis on effective integration of analysis, modeling, and simulation tools in transportation planning (Nevers et al., 2013). This makes it particularly valuable for developing and testing advanced transportation network management strategies, where users need to quickly visualize the impacts of different control policies or infrastructure modifications (Papageorgiou et al., 2003).

Multi-dimensional Matrix Representations

At the heart of NeXTA's analytical power is its sophisticated mathematical representation of traffic data. We capture agent movements using a trajectory tensor $X(a)$ that includes comprehensive information about each vehicle or traveler:

- Unique agent identity and classification
- Origin-destination pair information
- Detailed route specifications and path choices
- Precise departure and arrival timing
- Vehicle characteristics and driver behavior parameters

To support robust optimization and calibration procedures, we also maintain a baseline prior estimate $X_h(a)$ that provides regularization when working with noisy or incomplete data—a common challenge when dealing with real-world trajectory datasets. Complementing this individual-level data, measurement tensor $Y(i, j, t, m)$ tracks the real-time state of network links, capturing flow rates, traffic density, average speeds, and other performance metrics. The connection between individual trajectories and aggregate network conditions is established through proportional mapping operators that have their roots in classical traffic assignment theory (Beckmann et al., 1956; Wardrop & Whitehead, 1952).

Computational Graph for Traffic Modeling

The mathematical relationships underlying NeXTA draw from decades of traffic flow theory while incorporating modern computational approaches. One of NeXTA's most innovative features is its use of computational graphs to structure traffic modeling calculations, as illustrated in Figure Figure 1. This approach enables efficient forward and backward propagation of gradients, supporting optimization-based calibration of travel times and flows.

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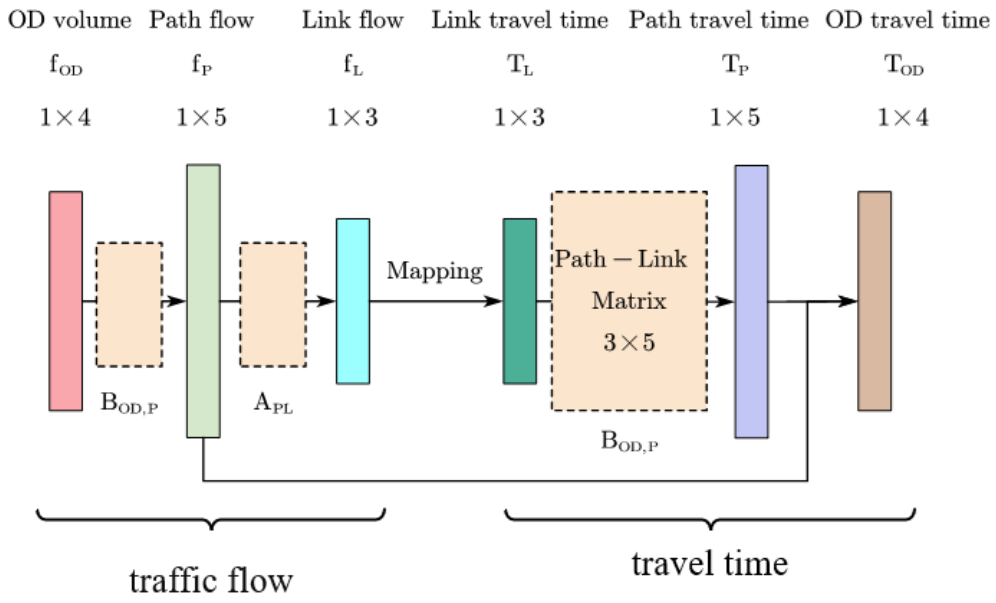


Figure 1: Layered computational graph representing OD-to-link relationships through path-based assignments and travel time transformations.

The multi-layer structure transforms data systematically between different network elements. In **Layer 1**, we convert origin-destination flows F_{OD} into path flows f_P using an OD-to-Path assignment matrix B: $f_P = B \cdot F_{OD}$. This step captures how travelers choose among alternative routes, incorporating principles from discrete choice theory (Ben-Akiva & Lerman, 1985; McFadden, 1974).

Layer 2 then aggregates these path flows to produce link flows f_L using a Path-to-Link incidence matrix A: $f_L = A \cdot f_P$. This is where individual route choices combine to create the traffic loads we observe on network links.

The reverse transformation happens in **Layer 3**, where link travel times T_L are aggregated into path travel times T_P using the transpose relationship: $T_P = A^\top \cdot T_L$. Finally, **Layer 4** maps these path travel times back to OD-level travel times using $T_{OD} = B \cdot T_P$.

This computational graph approach isn't just mathematically elegant—it's also highly practical. The structure naturally supports gradient-based optimization methods, making it much easier to calibrate complex traffic models against observed data.

The beauty of this layered approach becomes clear when you consider how it bridges abstract mathematical modeling with practical, user-friendly analysis tools. Every component of the computational graph shown in Figure Figure 1 maps directly to specific visualization features within NeXTA's interface, creating a seamless connection between theory and practice.

Table **??** shows exactly how this mapping works. Elements like OD flows, path flows, and travel times aren't just mathematical abstractions—they become interactive visual components that help users understand their transportation systems. This connection between rigorous modeling and intuitive visualization is what makes NeXTA particularly effective for model calibration, validation, and communication with stakeholders who may not have deep technical backgrounds.

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Table 2: Modeling Elements and Visualization Support in NeXTA {#tab:modeling_NeXTA}

Modeling Element	NeXTA Visualization Feature
OD Flows (F_{OD})	Interactive OD pair filtering and summary statistics
Path Flows (f_P)	Comprehensive route/path analysis view
Link Flows (f_L)	Real-time link performance view (flow, speed, density)
Link Travel Times (T_L)	Dynamic heatmaps and link-level metrics dashboard
Path Travel Times (T_P)	Aggregated trajectory visualizations with time profiles
OD Travel Times (T_{OD})	OD pair-level average travel time matrices
Tensor Representation ($X(a)$)	Multi-dimensional filtering and agent tracking

Practical Applications and Examples

OD-Based Trajectory Analysis

NeXTA offers a scientifically grounded environment for exploring transportation dynamics across spatial and temporal dimensions. At the core of this capability is the OD pair and path-based vehicle trajectory filtering interface, illustrated in Figure Figure 2. This tool allows users to dissect complex movement patterns by filtering trajectories based on origin-destination zones, departure time windows, and vehicle classifications—enabling focused analysis of freight activity, peak-hour congestion, or regional travel demand.

These filtering capabilities are tightly coupled with the platform's multi-layer analytical framework. Filtering by OD zones corresponds to **Layer 1**, where flows are interpreted in terms of traveler route choices across alternative paths. Once trajectories are narrowed, users can transition to **Layer 2** analyses, examining how those path-based movements translate into link-level congestion and flow conditions. In this view, NeXTA allows users to visualize how route-level decisions—aggregated from large-scale vehicle trajectories—influence link performance metrics such as volume, speed, and travel time.

The tool also supports downstream analyses aligned with **Layers 3 and 4**, enabling users to assess how link-level conditions affect cumulative travel experiences across entire paths and back to OD-level performance summaries. This layered perspective ensures that filtered trajectories are not only visualized, but also quantitatively analyzed within a consistent system that spans OD flows, path assignments, and link-level dynamics.

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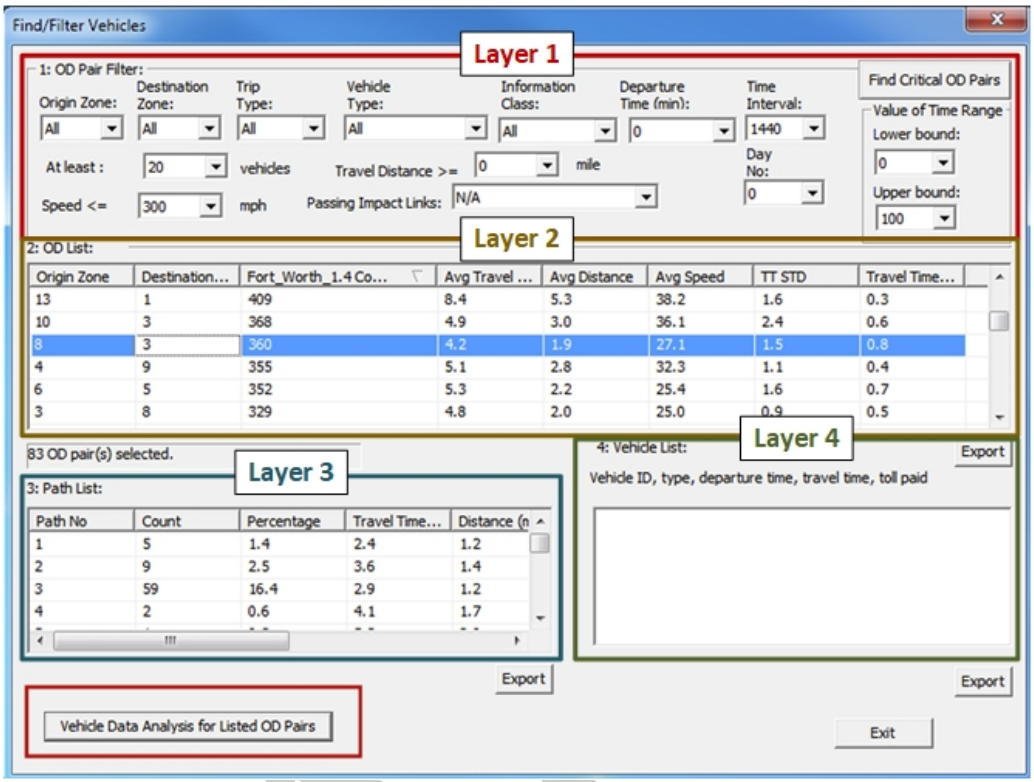


Figure 2: OD Pair and Path-based vehicle trajectory filtering in NeXTA, showing the interface for analyzing travel patterns across different origin-destination pairs.

Building on this structured framework, NeXTA's approach to route-based trajectory visualization aligns closely with established practices in the field—particularly those outlined in the FHWA's comprehensive *Trajectory Processor User's Guide* (Federal Highway Administration, 2011). At the same time, NeXTA advances these methods by offering a standardized GMNS-compliant environment that integrates OD-based filtering with path-level analysis and link-level performance metrics.

High-Resolution Empirical Data Analysis

Figure Figure 3 showcases NeXTA's capabilities when working with high-resolution empirical datasets. In this example, we're visualizing trajectories with the NGSIM I-101 dataset, focusing on Lane 2 with speed and spacing filters applied to extract vehicle behavior patterns within specific spatial and temporal constraints.

This type of detailed analysis opens up entirely new possibilities for understanding traffic flow phenomena. Users can study lane utilization patterns to optimize lane management strategies, identify stop-and-go wave propagation to understand congestion formation, and analyze acceleration profiles to calibrate car-following models. The visual rendering incorporates dynamic overlays—speed profiles, trajectory segment colors, density indicators—that make it much easier to spot important phenomena like shockwave formation, merging conflicts, or capacity bottlenecks.

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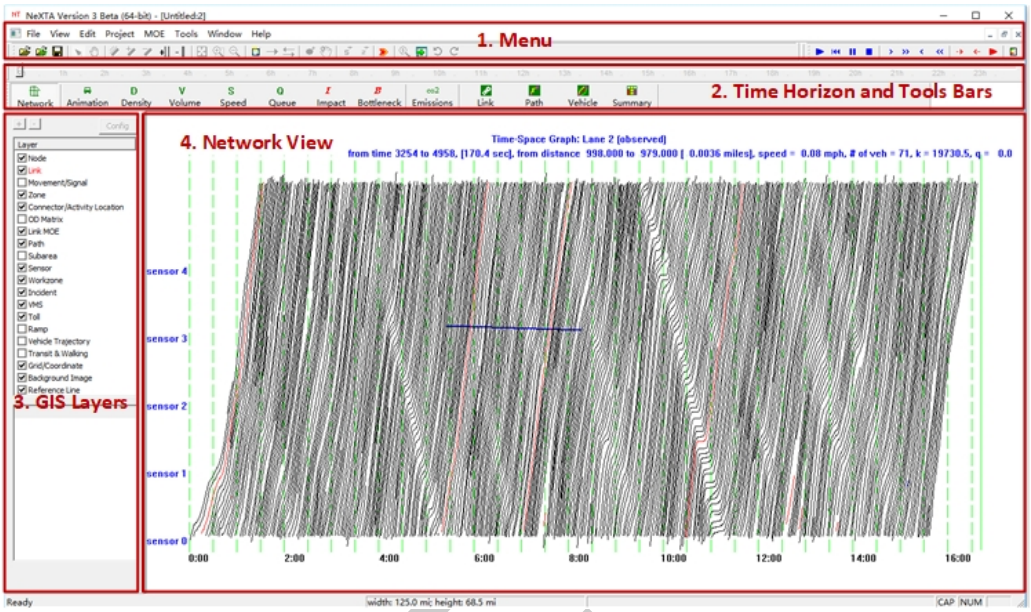


Figure 3: 2D NGSIM 101 Trajectory Visualization of Lane 2 with distance and speed selection filters applied to highlight specific vehicle behavior patterns.

This capability proves particularly valuable when you're trying to validate microsimulation results against observed data or calibrate model parameters based on real-world traffic dynamics. Traditional approaches to this kind of validation often rely on aggregate statistics that can mask important behavioral details. NeXTA's trajectory-level visualization allows for much more nuanced comparisons between observed and simulated behavior, making it an essential bridge between theoretical model development and practical deployment.

The integration with empirical datasets like NGSIM also supports more advanced applications, including the development of machine learning models for traffic prediction (Kim et al., 2020), the calibration of emerging connected and automated vehicle models (Shladover et al., 2012), and the validation of traffic flow theories under real-world conditions.

Research Impact Statement

NeXTA represents a significant step forward in making sophisticated transportation analysis accessible to a broader community of researchers and practitioners. By combining rigorous mathematical foundations with intuitive visualization interfaces, we've created a tool that bridges the gap between theoretical traffic modeling and practical transportation planning applications.

Looking ahead, we're excited about several emerging directions. The integration of machine learning approaches with traditional traffic flow theory offers promising opportunities for more accurate and robust transportation models (Zhang & Xie, 2011). The growing availability of connected vehicle data will create new possibilities for real-time network analysis and control (Talebpour & Mahmassani, 2016). And the increasing emphasis on multimodal transportation systems will require more sophisticated tools for analyzing the interactions between different travel modes (Ampountolas et al., 2017).

We believe NeXTA's flexible, extensible architecture positions it well to support these evolving needs while maintaining its core strengths in visualization and analysis.

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AI Usage Disclosure

Generative AI tools were used to improve sentence clarity and suggest phrasing in several sections of the paper. All technical content, mathematical formulations, software design decisions, and claims regarding functionality were authored, verified, and validated by the human authors. No AI-generated code was incorporated into the NeXTA software itself.

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